HW Responses

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HW10: 11/12

Herein, I respond to the articles on *Teaching Mathematics Graduate Students How to Teach* and *Math Anxiety in the Classroom.*

1. I think *experience* and *good judgement* are obvious qualities of effective teachers. I suppose I have a metaphysical disagreement with these as emphases for development; why did we choose them? The author says that one can acquire good judgement by analyzing and learning from personal experience or from a written case study, and I suspect we focus on the two key factors because they were discussed in some case study. However, I believe we would benefit from decomposing the subject matter into first principles, and I would feel much more comfortable emphasizing certain factors if we are able to prove that they deserve emphasis over any other factors.

2. I am conflicted about the author's "do and don't do" lists. Many good-hearted people aim to correct the injustice of gender-based discrimination by essentially ignoring gender, but I believe that's one of the many pitfalls of the trend of postmodernism. We should not be blind to the truth; ideally, we would ideally acknowledge it, and overcome it. I believe the author agrees, as they implement a corrective measure by advising positive role models of both sexes, but I believe that effort in sufficient.

   Basically, I think the gender-based achievement gap problem is not as clear as the author insists; I understand there are many achievement gaps with many consequences, but I don't understand which are problematic and which take priority. I'd be much more comfortable addressing their suggestions if I understood the their intended goals.

3. The authors address that many teachers experience difficulty facing students with different motivations. In particular, I still wonder how to deal with students who have no real interest in the subject matter, which is a problem much more severe and pervasive than is acknowledged.

   In the realm of calculus for instance, some things matter, some things don't, and some matter more than others. Teachers come to judge these things and impart that judgement on their students, who we hope reside in this realm of calculus. But many students don't live there, just as there are plenty of subjects I don't appreciate now or possibly ever. When it comes down to it, many students really shouldn't be taking calculus at all, if they won't appreciate it or otherwise benefit from it. From the perspective of technique, students outside the physical or applied sciences probably won't use it. Other skills can probably be acquired elsewhere, e.g. through classes on physics, computer science, or analytic philosophy. Even for students who would benefit from it, that benefit is essentially meaningless from a nihilistic point of view. While I don't think it's a particularly productive point of view, we are compelled to acknowledge because students (and people in general) do face it at times.

   I would like to learn how to balance the inherent meaninglessness of course material with motivations for studying it, both as a student and as a teacher.

   But I think it's hard to come by good arguments along those lines because authors have incentive to address their readers (i.e. "preach to the choir") who have difficulty acknowledging their own motivations.